

MAYUR PRADEEP KAMBLE

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PROFESSIONAL SUMMARY

Automation Test Engineer with 2+ years of experience in automotive embedded testing and Hardware-in-the-Loop (HIL) automation. Strong hands-on experience with Python, Pytest, Robot Framework, dSPACE AutomationDesk, ControlDesk, CAN/J1939 protocols, CAPL scripting, and Vector CANalyzer. Experienced in developing Python automation frameworks for functional and regression testing, reducing manual testing effort by approximately 40%. Skilled across requirement analysis, test planning, test case design, execution, defect management, STLC, and Agile/Scrum environments, with additional web/API automation experience.

TECHNICAL SKILLS

Programming & Scripting: Python, OOP, CAPL

Automation Frameworks: Pytest, Robot Framework, Selenium WebDriver, Unittest

Automotive & Embedded: Hardware-in-the-Loop (HIL), dSPACE AutomationDesk, dSPACE ControlDesk, CAN, J1939, Vector CANalyzer

Testing: Functional Testing, Regression Testing, Embedded Testing, HIL Testing, GUI Testing, Black Box Testing, STLC

Development & Tools: Git, Jira, Confluence, Polarion, PyCharm, Visual Studio

Computer Vision: OpenCV, Tesseract OCR, OpenPyXL

Methodologies: Agile, Scrum

PROFESSIONAL EXPERIENCE

L&T Technology Services

Associate Engineer – HIL Testing | 2024 – 2025

- Developed and maintained Python-based automation scripts for embedded systems and Hardware-in-the-Loop (HIL) testing using dSPACE AutomationDesk and ControlDesk.
- Developed Python automation frameworks for functional and regression testing, reducing manual testing effort by approximately 40%.
- Automated end-to-end HIL test workflows by integrating Python, Pytest, Robot Framework, Selenium WebDriver, and dSPACE environments.
- Designed and executed test cases for CAN and J1939 communication protocols using CAPL scripting and Vector CANalyzer.
- Developed OpenCV-based display and GUI validation utilities integrated with Tesseract OCR to automate text-rendering verification.
- Created real-time monitoring and reporting utilities to provide actionable testing insights during testing phases.
- Participated in requirement analysis, test planning, test case design, test execution, defect management, and STLC activities.

L&T Technology Services

Engineer – Web Automation | 2025 – Present

- Performed data load and integration testing for bulk data imports, validating error handling and database status updates.
- Validated API payloads and backend interactions by manipulating JSON/XML requests and verifying HTTP responses using Postman.
- Designed and maintained scalable automation architecture and migrated scripts from Robot Framework to Katalon and Pytest.
- Supported web and backend automation activities in Agile/Scrum environments.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering

Rajarambapu Institute of Technology, Sangli, Maharashtra | CGPA: 7.59 | 2017